



Grade 6 Mathematics

Compass · Ruler · Precision · Logic

Subject	Mathematics – Geometry
Grade	Grade 6
Topic	Perpendicular Bisector & Angle Bisector
Tools	Compass, Ruler, Pencil, Eraser
Duration	2 × 40-minute lessons

1. What Are Geometric Constructions?

Geometric constructions are drawings made using **only a compass and a straight ruler** — no measurements allowed! They help us create perfect shapes, lines, and angles with mathematical precision. Constructions teach us logical thinking and accuracy — skills used in art, architecture, engineering, and everyday problem-solving.

Learning Objectives

- Understand what a geometric construction is and why measurements are not used.
- Identify and correctly use a compass, ruler, pencil, and eraser.
- Construct a **perpendicular bisector** of a given line segment step by step.

- Construct an **angle bisector** of a given angle step by step.
- Compare the two types of bisectors and explain how they differ.
- Recognise real-life applications of bisectors.

2. Tools for Constructions

Tool	Purpose	Tips for Use
Compass	Draws circles and arcs of a fixed radius	Keep the compass width constant between steps
Ruler (Straightedge)	Draws straight lines	Use only for drawing — not for measuring
Sharp Pencil	Makes precise, fine marks	A sharp HB pencil gives the best accuracy
Eraser	Removes construction marks when needed	Lightly erase only unwanted arcs

3. Perpendicular Bisector

Definition: A **perpendicular bisector** is a line that cuts a given line segment into **two equal parts at exactly 90°** . It passes through the midpoint of the segment and is perpendicular to it.

Key Properties

- Divides the segment into two congruent (equal) halves.
- Meets the segment at a right angle (90°).
- Every point on the perpendicular bisector is equidistant from both endpoints.

Step-by-Step Construction

Step 1

Draw the Line Segment

Draw line segment AB of the given length using a ruler and pencil. Label the two endpoints clearly as A and B.

Step 2	Arc from Point A Place the compass point at A. Open the compass to more than half the length of AB. Draw two arcs — one above and one below the line segment. Keep the compass width unchanged.
Step 3	Arc from Point B Without changing the compass width, place the compass point at B. Draw two more arcs above and below the line. These arcs must cross the arcs drawn from A, creating two intersection points (call them P and Q).
Step 4	Draw the Bisector Use a ruler to draw a straight line through P and Q. This line PQ is the perpendicular bisector of AB. It passes through the midpoint M of AB at a 90° angle.

— Practice Exercises – Perpendicular Bisector

- ★ Construct the perpendicular bisector of line segment AB = 8 cm.
- ★ Construct the perpendicular bisector of line segment PQ = 6 cm.
- ★ Construct the perpendicular bisector of line segment XY = 10 cm.
- ★ Challenge: Draw any triangle. Construct perpendicular bisectors of all three sides. What do you notice about where they meet?

4. Angle Bisector

Definition: An **angle bisector** is a ray that divides an angle into **two equal smaller angles**. It starts from the vertex and passes through the interior of the angle.

Key Properties

- Divides the angle into two congruent (equal) parts.
- Starts at the vertex of the angle.
- Every point on the angle bisector is equidistant from both rays of the angle.

Step-by-Step Construction

Step 1	Draw the Angle Draw angle ABC with the vertex at B. Label the vertex B and the two rays BA and BC clearly.
Step 2	Arc from the Vertex Place the compass point at the vertex B. Draw an arc that crosses both rays of the angle. Label the intersection points D (on ray BA) and E (on ray BC).
Step 3	Arcs from D and E Place the compass point at D and draw an arc inside the angle. Without changing the compass width, place the compass at E and draw another arc. The two arcs must intersect — label this point F.
Step 4	Draw the Bisector Ray Draw a ray from vertex B through point F. Ray BF is the angle bisector of angle ABC. It divides the angle into two equal halves: $\angle ABF = \angle FBC$.

— Practice Exercises – Angle Bisector

★ Construct the angle bisector of a 60° angle.
★ Construct the angle bisector of a 90° angle.
★ Construct the angle bisector of a 120° angle.
★ Challenge: Draw any triangle. Construct angle bisectors from all three vertices. Where do they meet?

5. Comparison: Perpendicular Bisector vs Angle Bisector

Feature	Perpendicular Bisector	Angle Bisector
What it bisects	A line segment	An angle
Result	Two equal halves of the segment	Two equal smaller angles
Angle formed	Always 90°	Depends on original angle

Type of result	A line (or line segment)	A ray
Starting point	Midpoint of segment	Vertex of angle
Common tool	Compass and ruler	Compass and ruler

6. Real-Life Applications of Bisectors

Bisectors are not just textbook concepts — they are used in the real world every day!

- **Architecture:** Architects use perpendicular bisectors to design symmetrical buildings and ensure structural balance.
- **Engineering:** Engineers apply bisector principles when building bridges, roads, and machine components that must be perfectly aligned.
- **Art and Design:** Artists use bisectors to create symmetrical patterns, logos, and decorative designs.
- **Navigation:** Navigators use bisector principles to find equidistant points and plot accurate courses on maps and charts.
- **Computer Graphics:** Software for 3D modelling and animation uses bisector algorithms to calculate normals and reflections.

7. Summary & Key Takeaways

✓	A perpendicular bisector divides a line segment into two equal parts at 90° .
✓	An angle bisector divides an angle into two equal parts.
✓	Both constructions use only a compass and ruler — no measurements.
✓	The key to accurate constructions is keeping the compass width constant .
✓	Bisectors have important uses in architecture, engineering, art, and navigation.

Homework: Complete all six practice exercises in your notebook. For each construction, show all arcs and label every point clearly. Bring your work to the next class for review.

